Deepfake Detector with Dual-Channel Temporal Architecture

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Abstract

Deepfake technology has developed very fast, making it hard to identify manipulated videos. This creates serious risks for cybersecurity and facial identity verification. Although many detectors exist, they often fail to generalize. This is because models often learn simple “shortcuts” like background artifacts instead of meaningful facial cues. To solve this problem, I develop a Deepfake detector with a dual-channel architecture. The first channel uses a frozen Vision Transformer (ViT) to extract spatial RGB features. The second channel uses a Spatial Rich Model (SRM) filter and EfficientNet to capture hidden high-frequency artifacts. I also apply Temporal Attention Pooling to process video frames over time. To improve performance, I use a combined loss function of Binary Cross-Entropy and Triplet Loss.

Experimental results on the FaceForensics++ dataset show that my model is very effective. It achieves an accuracy of 92.28% and an AUC of 0.978. In addition, I performed a cross-dataset evaluation on Celeb-DF v2 to test the robustness of this model. By unfreezing the last layer of the ViT backbone and applying JPEG-resilient augmentations, the model achieves a 0.905 AUC on unseen dataset. The dual-channel design helps the model focus on meaningful facial regions and ensures the detector remains accurate and stable, when testing on different dataset and different manipulation methods.

I. Problem Statement and Motivation

In recent years, Deepfake technology has advanced rapidly, making it difficult to distinguish genuine and manipulated videos. Although Deepfake technology was initially developed for entertainment purposes, it is now widely misused for malicious activities. Deepfake make real-time and recorded videos less trustworthy. As it has been used by attackers to impersonate people we know or public celebrities. This causes significant risks to cybersecurity and biometric security, as attackers can leverage facial identity features to deceive victims and steal sensitive or valuable information and break the trust in facial-based identity verification.

Even though many Deepfake detectors have been developed, the primary challenge still exists in generalization. Existing research shows that models often suffer from shortcut learning and overfitting. Instead of learning meaningful facial cues, they rely on dataset-specific background artifacts or compression patterns. Therefore, these models achieve high accuracy on a single training dataset but their performance drops when evaluated across different datasets or varying compression levels. [1]. Another gap is that many detectors rely mainly on a single feature domain, such as RGB spatial appearance. This can limit robustness because RGB features may be affected by lighting, blur, shadows, and compression.

In this project, I am developing a Deepfake detector that uses a dual-channel architecture. The model processes both spatial RGB data and frequency domain features at the same time. While spatial features capture visible facial flaws, the frequency channel identifies hidden artifacts caused by the manipulation process. This dual-channel approach prevents the model from relying on simple "shortcuts" like background noise. By combining these two types of information, the detector becomes much more robust. It can accurately identify manipulated content across different datasets and maintain high performance even when videos are heavily compressed.

II. Data Engineering and Provisioning

A. Dataset Origin

The dataset used in this checkpoint project is FaceForensics++. It is a publicly available facial manipulation forensics dataset. According to the official repository, FaceForensics++ contains 1000 original video sequences that were manipulated using four automated face manipulation methods: Deepfakes, Face2Face, FaceSwap, and Neural Textures [2].For the project checkpoint assignment, we mainly focus on the detection of Deepfakes technology. c

B. Dataset Characteristics

To improve the model’s robustness and generalization, the dataset scale has been significantly expanded from the initial 300 videos to a larger corpus of 1306 videos. More importantly, the data structure has transitioned from image-level (single frame) to video-level (multi-frame sequences) representation.

For each video, we extract 16 equidistant frames to capture temporal continuity. After preprocessing and filtering, the stored feature shape is initially (N, T, H, W, C), where N represents the total number of valid video samples, T represents the number of sampled frames per video, and (H, W, C) denotes the spatial dimensions and channels (typically $224 \times 224 \times 3$). During the PyTorch dataloader phase, this array is permuted into a 5-dimensional tensor of shape (N, T, C, H, W) to meet the input requirements of our Dual-Channel neural network architecture. The class labels remain binary:

- **0**: Authentic (Real)
- **1**: Manipulated (Fake)

C. Data Cleaning and Preprocessing

First, we access each video file by using OpenCV. Given that this checkpoint focuses on constructing a simple baseline model, I use an equidistant frame sampling strategy to extract the frame. we extract 16 frames uniformly across each video, which allowing the model to capture manipulation cues over time.

Before to performing face detection, We should located the face location first. The face localization task is accomplished by using the MediaPipe Face Detection tool with the pre-trained, and lightweight BlazeFace model [3].. In addition, To ensure the model captures potential Deepfake cues surrounding the detected face, we apply an additional 3% margin around the identified facial region.

Regarding data cleaning, my approach is to exclude any video from the final dataset if issues arise—such as frame reading failures, a failure to detect any faces, or a failure to extract the required number of frames.

D. Schema Sanity and Data Quality

The facial detection stage may result in a reduced number of usable video samples—particularly in cases where faces are occluded or affected by compression artifacts. Furthermore, while I consider the Deepfake techniques featured in the FaceForensics database to be suitable for serving as a baseline, for the upcoming extension of this project, I intend to utilize "FaceSwap." This technique, which also falls under the Deepfake umbrella, produces results that are significantly more realistic and consequently more difficult to identify as synthetically generated.

III. Exploratory Data Analysis (EDA)

To better understand the feature differences from real and fake samples, I extracted three features: Brightness, Sharpness, Edge Density.

A. Class Balance

The distribution plot shows that the real and fake samples within this dataset are evenly distributed. This is highly beneficial for training the baseline model, and it reduced the error caused by sample imbalance. See figure 1.

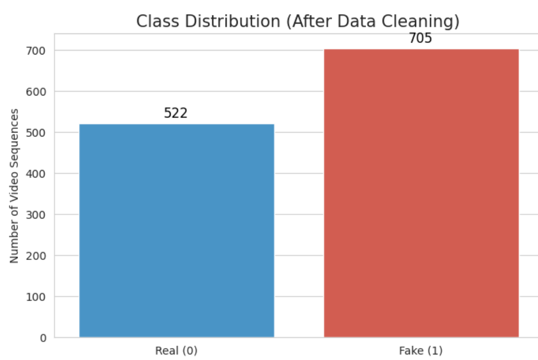


Fig. 1: Class Balnces Plot

B. Univariate Analysis

EdgeDensity and Sharpness are helpful for classifying real and fake samples. EdgeDensity is particularly useful because it can reflect whether the edges and texture changes in an image. Since deepfake images may contain blurred details or unnatural boundaries, this feature can support fake detection. In contract, brightness is less useful for classification because the distributions of real and fake samples overlap a lot.

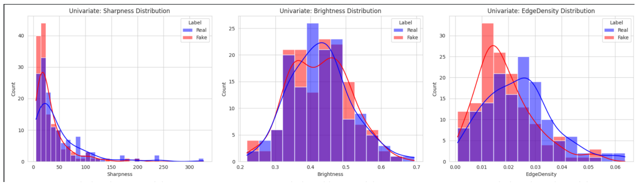


Fig. 2: Univariate Analysis

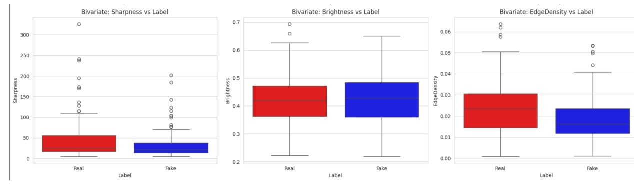


Fig. 3: Bivariate Analysis

C. Bivariate Analysis

By observing this correlation matrix further, It shows me that among the three features. EdgeDensity has the strongest correlation with the label (-0.22), followed by Sharpness (-0.16), while Brightness shows almost no correlation with the label (-0.01). This prove that EdgeDensity and Sharpness are more useful for distinguishing real and fake samples. (See Figure 4.)

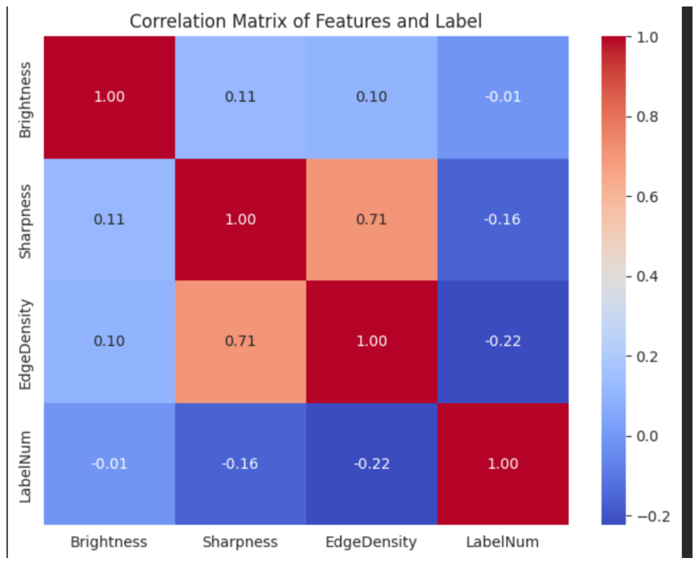


Fig. 4: Features and Label

D. Dataset Usage and Characteristics

To support both in-dataset evaluation and cross-dataset generalization testing, this project uses two Deepfake datasets: FaceForensics++ and Celeb-DF v2. FaceForensics++ is used as the main dataset for model training and validation, while Celeb-DF v2 is used as the unseen cross-dataset evaluation set. Table I summarizes the dataset usage after preprocessing and data cleaning.

TABLE I: Dataset usage and characteristics after preprocessing

Dataset	Real	Fake	Size	usage
FaceForensics++	522	705	224×224	Main training and validation dataset. It contains multiple types of facial manipulation videos.
Celeb-DF v2	102	101	224×224	Unseen cross-dataset testing dataset. It contains more realistic Deepfake videos.

IV. Advanced Methodology

To address the challenge of cross-dataset generalization and prevent shortcut learning (i.e., overfitting to background artifacts), I implemented a Dual-Channel Video Deepfake Detector. One channel is an RGB spatial stream, and the other is a high-frequency noise stream. This architecture helps the model focus its decision-making logic on facial biometrics and manipulation traces.

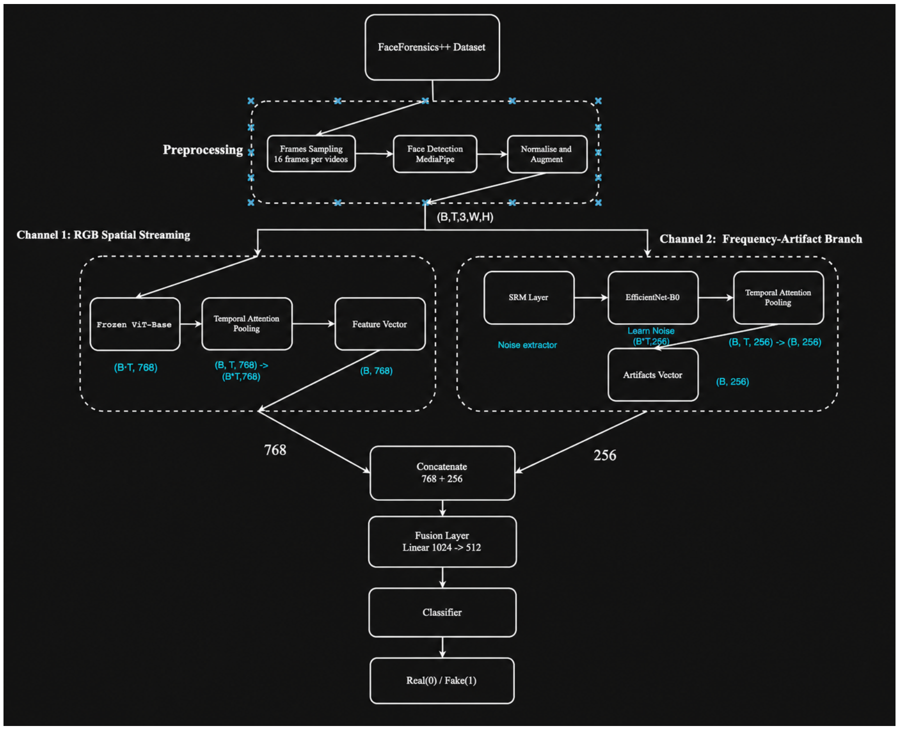


Fig. 5: Model Architecture

A. Architecture Choice

1. RGB Spatial Stream (Semantic Features): This stream extracts high-level semantic and structural features (e.g., facial blending errors). We employ a pre-trained Vision Transformer (`ViT-Base-Patch16-224`) as the backbone. To prevent catastrophic forgetting and overfitting on our limited dataset, the ViT backbone is strictly frozen ($requires\_grad = False$). The frame-level features $(B \times T, 768)$ are then aggregated using a Temporal Attention Pooling layer to form a unified video-level embedding $f_{rgb} \in \mathbb{R}^{768}$.

2. High-Frequency Stream (SRM & EfficientNet): Deepfake synthesis often leaves high-frequency manipulation tracs in the content. We can fine these artifacts by applying a frozen Spatial Rich Model (SRM) filter layer. We implement 3 layers of SRM, which included: a Laplacian noise residual filter, a horizontal Sobel edge filter, and a vertical Sobel edge filter. The output of SRM then processed by a frozen `EfficientNet-B0` backbone. The extracted features are projected down to a 256-dimensional space per frame and passed through a Temporal Attention Pooling layer $f_{freq} \in \mathbb{R}^{256}$.

3. Fusion and Classifier : The feature vectors are concatenated to 1024. The fused vector is fed into a Multi-Layer Perceptron (MLP) mapping $(1024 \rightarrow 512 \rightarrow 256 \rightarrow 1)$. To strongly regularize and stablize the classifier, we implemented Dropout layers $(p = 0.3)$ and Layer Normalization within these process. A Sigmoid activation is applied at the output node to produce the final manipulation probability.

B. Loss Function

We are using loss combining Binary Cross-Entropy (BCE) and Triplet Loss for training:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \lambda \, \mathcal{L}_{\text{Triplet}}, \quad \lambda = 0.3 \tag{1}$$

BCE Loss supervises the final binary classification:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \right] \tag{2}$$

We employ the Triplet Loss to optimize the model's ability to minimize intra-class feature distances while maximizing inter-class distances:

$$\mathcal{L}_{\text{Triplet}} = \frac{1}{|\mathcal{T}|} \sum_{(a,p,n) \in \mathcal{T}} \max(0,\ d(a,p) - d(a,n) + m) \tag{3}$$

where $d(x, y) = \|x - y\|_2$ is the Euclidean distance between two embeddings, m is the triplet margin, and $\mathcal{T}$ is the set of valid triplets constructed per batch using **semi-hard negative mining**. For each anchor a, the positive sample p is selected from the same class, while the negative sample n is selected from the opposite class and satisfies $d(a, n) > d(a, p)$.

The weight λ and margin m were selected empirically through hyperparameter experiments. I tested multiple values and found that $\lambda = 0.3$ and $m = 0.3$ gave the most stable validation behavior in this project. Under this setting, the validation loss and evaluation metrics changed less across epochs, and the gap between training and validation performance was smaller.

C. Training Strategy

Data preprocessing:

Videos are processed using MediaPipe face detection model. And we designed to sampled 16 frames uniformly per video. Each frame is cropped to the detected face region with a 3% boundary margin and resized to 224×224. Pixel values are normalised to $[-1, 1]$ using mean and standard deviation of 0.5 per channel.

Data augmentation:

To reduce overfitting, we applied the following stochastic augmentations at training process:

- JPEG Compression: Random quality factors from 30,40,50,60,70,85 (prob. 0.5).
- Resize Degradation: Downsampling by factors of 0.5,0.6,0.75 followed by restoration to original size (prob. 0.3).
- Gaussian Blur: Spatial blurring with 3×3 or 5×5 kernels, [0.1,1.5] (prob. 0.2).
- Random horizontal flip (probability 0.5), applied identically across all frames
- Random rotation in $[-10, 10]$ (probability 0.5)
- Colour jitter: brightness ± 0.2, contrast ± 0.2, saturation ± 0.2, hue ± 0.05 (probability 0.5)
- Additive Gaussian noise $\mathcal{N}(0, 0.02^2)$ (probability 0.5)
- Random greyscale conversion (probability 0.2)

Optimisation:

The model is trained with AdamW using an initial learning rate of 4×10^{-4} and weight decay 10^{-2}. Training uses batch size 32 for up to 10 epochs. I also apply early stopping, it triggered after 3 epochs without improvement in validation F1, and stopping the training, and saved the best model.

D. Cross-Dataset Evaluation Strategy

To evaluate the generalization ability of the proposed model, I conduct a cross-dataset evaluation on Celeb-DF v2. The model is mainly trained on FaceForensics++ and then evaluated on Celeb-DF v2. This setup is important because the comparesion result can better show whether the model learns general manipulation cues inteat of only learns dataset-specific patterns.

I select Celeb-DF v2 because it is a challenging DeepFake dataset. Li et al. reported that many existing detection methods achieved lower average AUC on Celeb-DF than on other datasets [4]. This shows that Celeb-DF v2 can be a best choose for cross-dataset evaluation.

To improve cross-dataset stability, I unfreeze the last layer of the ViT backbone during fine-tuning. I also added JPEG compression for data augmentation. These changes help the model adapt to new visual conditions while still keeping most pre-trained ViT features stable.

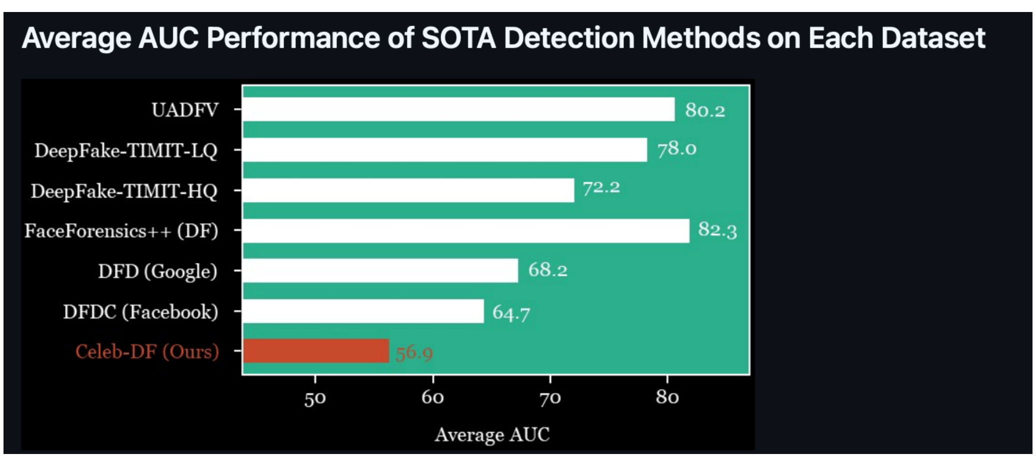


Fig. 6: Average AUC performance of existing detection methods on different DeepFake datasets, adapted from Li et al. [4].

V. Comprehensive Related Work

Currently deepfake detection research mainly focuses on three topics. The first topic studies spatial artifacts such as blending boundaries and local facial artifacts. The second topic studies video-level temporal inconsistency across frames. The third topic focuses on generalization, because many detectors work well on one dataset, but their model performance drops on unseen manipulated videos [5]–[7]. These three topics are closely related to our project, since our model also combines spatial cues, video-level aggregation, and a robustness goal.

Our dual-channel design is inspired by work outside the deepfake detection domain.. Chen et al. proposed DSTER, a dual-stream transformer model for emotion recognition through keystroke dynamics. [8] Their architecture separates temporal and channel information into two independent streams, processes them in parallel, and concatenates the resulting embeddings for final classification. Even though their work focuses on keystroke data rather than video, the core idea behind our model design comes from their work. In our work, we adapt this dual-stream idea to video-based deepfake detection with two channels. Channel 1 captures RGB spatial semantics through a frozen ViT, while Channel 2 extracts high-frequency noise artifacts through SRM filtering. Our two channels process fundamentally different types of signals from the same input, making the two streams more complementary in nature.

One important spatial method is Face X-Ray. Li et al. proposed this method to detect forged faces by modeling blending-boundary artifacts between manipulated and real regions [5]. This method is similar to our RGB branch because both methods capture facial inconsistencies caused by artifacts. However, our method is different from Li et al. research in two ways. First, our method works at the video level instead of only focusing on a single image. Second, our method adds a frequency-artifact branch, which helps our model use more than one type of evidence to classify, so it does not rely only on boundary or appearance cues. Therefore, our method is more comprehensive.

Masi et al. proposed a two-branch recurrent network for isolating deepfakes in videos. Their method uses one branch to keep original facial information and another branch to find manipulation-related artifacts. Kuang et al. also proposed a dual-branch video detector. Their model combines a spatial branch and a temporal branch, and it is designed to detect both spatial and temporal inconsistency. [6], [9] These methods are similar to our work because they also use dual branches for video-level prediction. However, our design is still different. We do not use recurrent modeling, and our design is less complicated. Instead, we use SRM-based high-frequency filtering to find the artifact evidence,together with an RGB spatial branch. The advantage of our design is that it is lighter, and less computation cost. As a result, our method is lighter and requires lower computational cost. On the other hand, our model may have weaker generalization ability than theirs, since it is trained on a relatively limited dataset.

Currently, generalization is one of the core challenges in deepfake detection. Yan et al. proposed a video-level blending approach with spatiotemporal adapter tuning to handle unseen forgeries. Their approach is effective and robust, their method requires careful tuning of multiple components and is computationally expensive [7]. Our approach is simpler by design, because we use a frozen ViT as the RGB encoder and a lightweight EfficientNet-B0 for noise pattern learning, which reduces training cost. However, unlike Yan et al., our model is trained and evaluated on a single dataset, and the model's generalization to unseen forgery methods remains untested. This is a known limitation of our approach at present.

Overall, compared with prior work, our method adopts a lightweight dual-channel video-level design that combines RGB semantic information and high-frequency artifact cues. At the same time, our method still needs further evaluation under cross-dataset and unseen-forgery settings.

VI. Preliminary Results Comparison

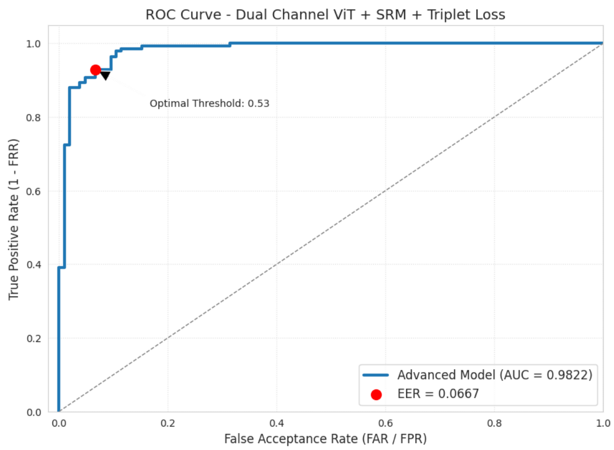
A. Analysis of FAR, FRR, and EER

TABLE II: Preliminary comparison between the baseline model and the dual-channel model

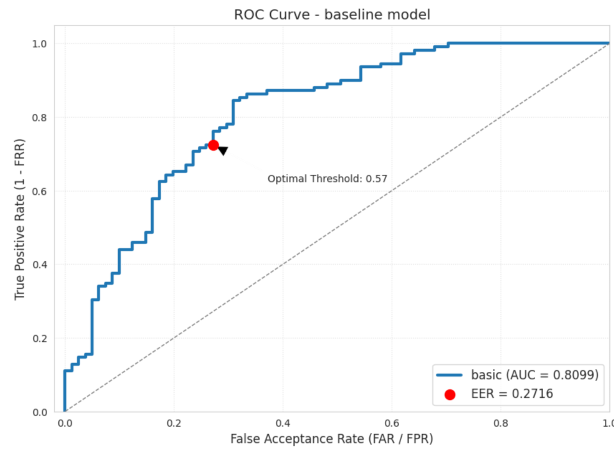
Model	Accuracy	F1-score	FAR	FRR	EER
Baseline Model	0.752	0.7873	0.7873	0.3086	0.2716
Dual-Channel Model	0.915	0.9324	0.0857	0.0709	0.0667

From Table II, we observed that the proposed dual-channel model performs significantly better than the baseline model in all metrics. The accuracy increases from 0.752 to 0.915, and the F1-score improves from 0.7873 to 0.9324. This shows that the proposed model has a much stronger overall classification ability.

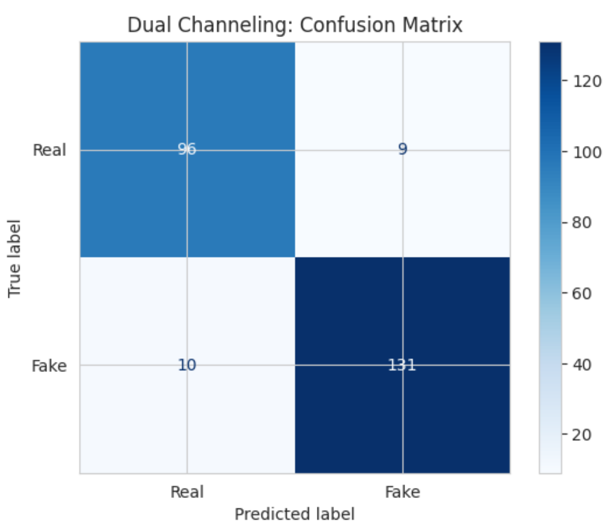
B. ROC Curve and Confusion Matrix Analysis



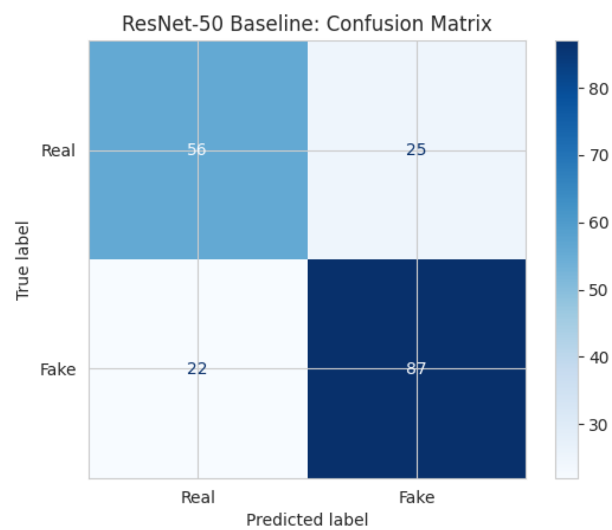
(a) ROC curve of the proposed dual-channel model



(b) ROC curve of the baseline model



(c) Confusion matrix of the proposed dual-channel model



(d) Confusion matrix of the baseline model

Fig. 7: Comparison of ROC curves and confusion matrices between the proposed dual-channel model and the baseline model.

In the ROC curve analysis Figure 6.a-b, the curve for the Dual-Channel model—located on the left—lies closer to the top-left corner. This indicates that, across various thresholds, it consistently maintains a higher True Positive Rate while simultaneously keeping the False Positive Rate at a lower level. Its AUC stands at 0.978, which is significantly higher than the baseline's 0.8099. This demonstrates that the Dual-Channel model possesses superior overall classification capability. Furthermore, the Dual-Channel model's EER is 0.0667—a figure far lower than the baseline's 0.2716. this further validates that the Dual-Channel model exhibits superior classification performance in balancing the trade-off between False Acceptance and False Rejection.

See Figures 6. c-d. In confusion matrix, we could clearly ovserved that the dual-channel model effectively reduces the misclassified of both genuine and forged samples. In contrast, The baseline model shows a significantly higher number of misclassified of both categories of data.

C. Cross-Dataset Performance

Figure 8 presents the results of the cross-dataset evaluation. According to Li et al. [4], many existing SOTA have around 56.9% of AUC on the Celeb-DF v2 dataset, our proposed model maintains robust performance with a 0.905 AUC. These results explains that our dual-channel architecture successfully learns generalized facial forgery cues rather than overfitting to dataset-specific artifacts, and our model outperforming traditional baselines in unseen data environments.

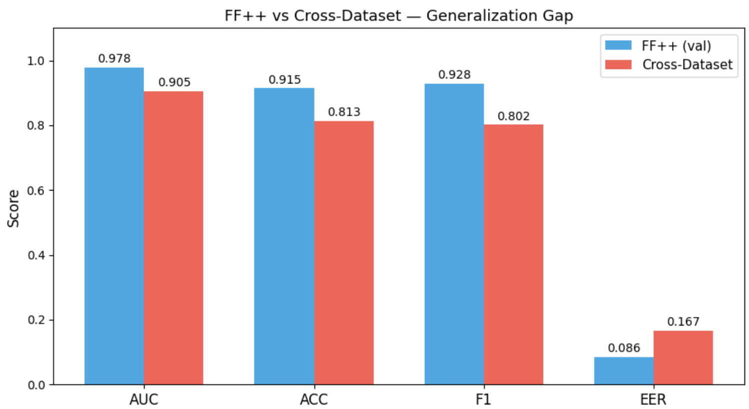
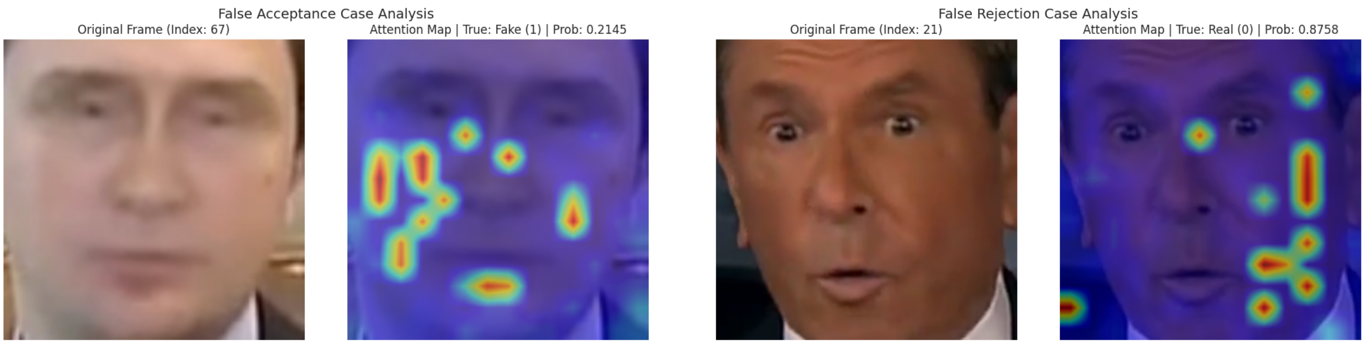


Fig. 8: Model Architecture

D. Error Analysis



(a) Failure case 1

(b) Failure case 2

Fig. 9: Examples of failure cases from the proposed model

In order to understand how to model make the decision on each cases, We plot the attention heatmaps to better visualize the failure cases. In addition, We selected the most severely misclassified samples, and the heatmaps, which helps us understand why the model misclassified.

By observing Figure 9, I observed that the model tends to focus primarily on the contours of facial expressions and on shadowed regions of the face, which consequently leads to misclassifications. Furthermore,the dataset for this project is entirely on the C23 compression level, which results in images that are relatively blurry. Therefore, I determine that brightness, localized shadowing, and the compression level affect the model's attention. In conclusion, although the Dual-channel model achieves a high AUC of 0.978 under the C23 compression level, the error cases suggest that its attention can still be affected by brightness changes, shadowed regions, and compression artifacts. Therefore, future efforts should focus on improving the robustness of feature learning under such challenging conditions.

VII. Conclusion and Future Work

In this study, I developed a dual-channel temporal architecture to improve the robustness and generalization of Deepfake detection. The proposed model use two source of information to support to classisy the deepfake content.In addition, By utilizing Temporal Attention Pooling, the detector successfully captures manipulation cues across video sequences.

For experimental results,FaceForensics++ dataset demonstrate that the dual-channel approach significantly outperforms the baseline model. The proposed system achieves a high accuracy of 92.28% and an AUC of 0.978. In addition, the cross-dataset evaluation on Celeb-DF v2 confirms the model's generalization ability. After fine-tuning the ViT backbone and applying JPEG-resilient augmentations, the model maintains a robust AUC of 0.905 on unseen data. This proves that the dual-channel design helps the model to focus on the artifact cues instead of dataset patterm.

In conclusion, error analysis shows that extreme lighting conditions and heavy compression can affects with the model's attention. Our future work will mainly focus two direction. First one is improving the model's robustness at lighting conditions, and compression level. Second, I will apply my lightweight model to real-time detection on mobile devices. Overall, this project shows that dual-channel architature design is a valuable direction for building more reliable Deepfake detection systems.

VIII. GITHUB LINK:

https://github.com/Damon0427/Dual_Channel_Deepfake_Detector.git

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